

California Highway Safety, 2024

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Introduction

California's 58 counties experience very different levels of highway traffic. Because of this, raw crash totals do not always tell the same story as crash rates. A county with heavy traffic may have thousands of crashes simply because more vehicles travel through it, while a county with much less traffic may have a higher crash rate relative to the amount of travel.

This report examines county-level 2024 highway safety patterns using crash counts, travel exposure, fatality measures, and selected pedestrian fatality counts, with bicycle fatalities included as a supplemental analysis. The analysis focuses on four main questions:

1. Which counties have the highest total crash rates relative to vehicle travel?
2. Which counties have the highest fatal-injury crash rates and fatality rates?
3. How do exposure-adjusted rates differ from absolute crash burden?
4. Where are pedestrian fatalities concentrated in absolute terms?

Data

The analysis uses California Open Data / Caltrans crash-summary data for the 2024 state highway system, supplemented with county-level pedestrian and bicycle fatality data. County boundaries used in ArcGIS Pro were derived from the U.S. Census Bureau's 2024 TIGER/Line county geography.

The cleaned analytical table contains one row for each of California's 58 counties and includes road miles, travel exposure, crash counts, victim counts, three exposure-adjusted safety rates, and pedestrian/bicycle fatality and injury counts.

Methods

Three primary safety measures are analyzed as exposure-adjusted rates:

- **Total Crash Rate:** Total crashes per million vehicle miles traveled (VMT).
- **Fatal-Injury Crash Rate:** Crashes involving a fatality or injury per million vehicle miles traveled.
- **Fatality Rate:** Fatalities per 100 million vehicle miles traveled (VMT).

Pedestrian and bicycle fatalities are treated separately as **raw county counts**, not exposure-adjusted risk measures.

For the cartographic analysis, the prepared county table was joined to 2024 Census county boundaries in ArcGIS Pro. Each thematic map uses a five-class graduated-color scheme based on Natural Breaks (Jenks). The same statewide extent and layout template were retained across maps for consistency.

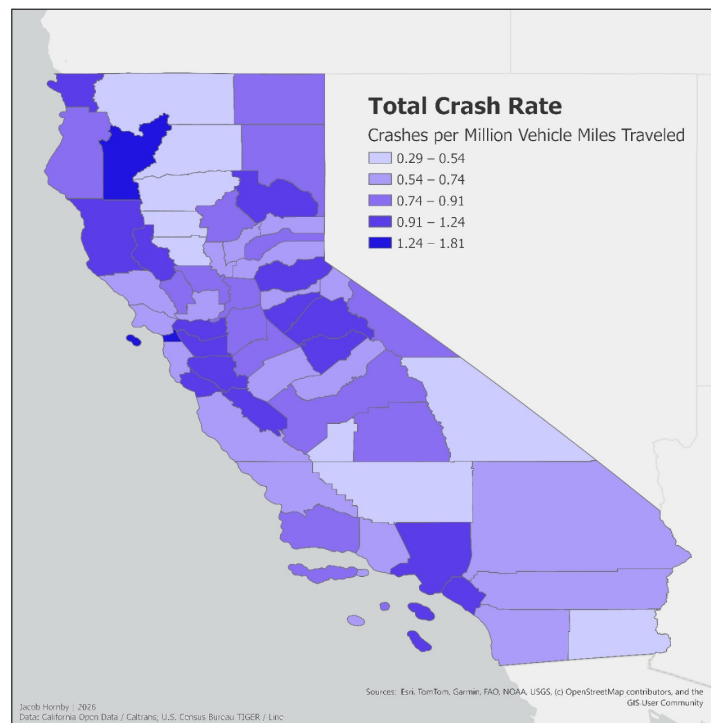
Results

Total Crash Rate

Top 10 California Counties by Total Crash Rate, 2024

County	Crash Rate	Total Number of Crashes	Vehicle Miles Traveled (Millions)
Trinity	1.813	208	114.7
San Francisco	1.523	1,884	1,237.2
Calaveras	1.236	330	266.9
Alameda	1.097	9,453	8,617.0
Tuolumne	1.091	325	298.0
Los Angeles	1.074	41,235	38,389.2
San Benito	1.058	379	358.3
Lake	1.049	415	395.7
Santa Cruz	1.035	1,018	983.5
Plumas	1.035	154	148.8

Total Crash Rate by California County, 2024



The total crash-rate map highlights counties where crash occurrence is high relative to vehicle travel. This differs from a raw crash-count map, which would be dominated by California's highest-volume counties.

Trinity County had the highest total crash rate in 2024 despite having relatively few crashes and comparatively little highway travel. San Francisco also ranked near the top. Los Angeles provides an important

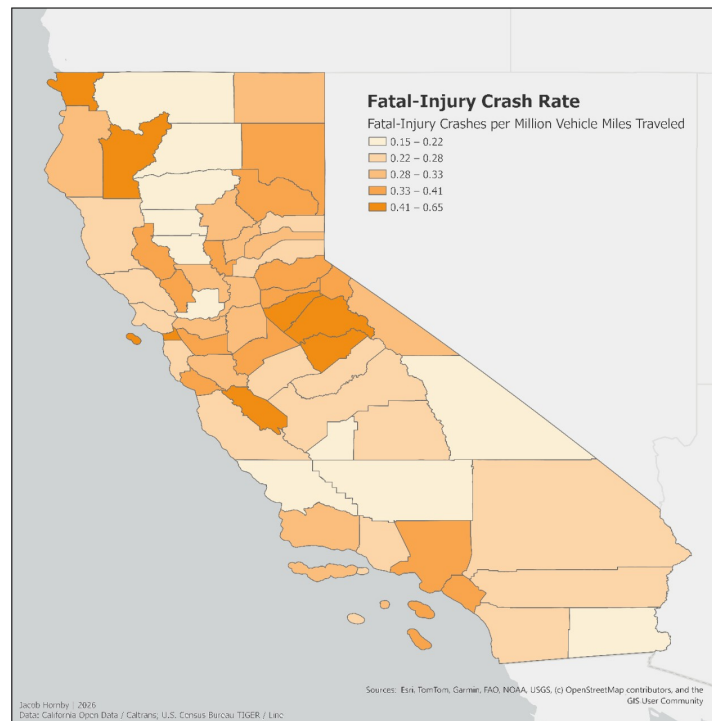
contrast: it had by far the largest number of crashes, but its crash rate was much less extreme once travel volume was taken into account.

Fatal-Injury Crash Rate

Top 10 California Counties by Fatal-Injury Crash Rate, 2024

County	Fatal-Injury Crash Rate	Injury Crashes	Fatal Crashes	Travel (Million VMT)
Trinity	0.654	71	4	114.7
Calaveras	0.521	132	7	266.9
Tuolumne	0.453	131	4	298.0
San Francisco	0.449	546	10	1,237.2
San Benito	0.438	151	6	358.3
Mariposa	0.433	51	2	122.3
Del Norte	0.427	74	4	182.5
Lake	0.407	145	16	395.7
Santa Cruz	0.404	391	6	983.5
Plumas	0.390	57	1	148.8

Fatal-Injury Crash Rate by California County, 2024



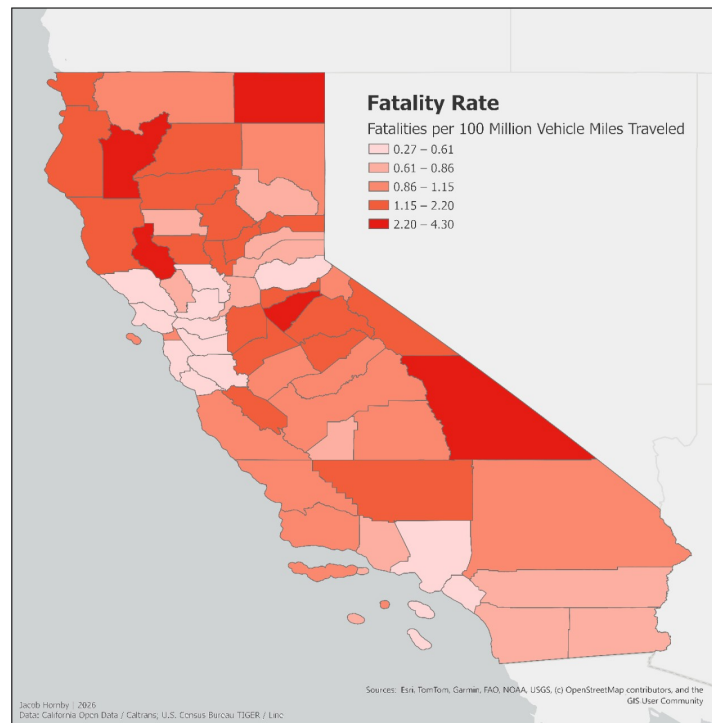
Trinity and Calaveras stand out again, suggesting that their elevated crash rates are not limited to lower-severity incidents. Tuolumne and San Benito also rank among the five highest counties for fatal-injury crash rates, while Lake remains within the top ten.

Fatality Rate

Top 10 California Counties by Fatality Rate, 2024

County	Fatality Rate	Fatalities	Fatal Crashes	Travel (Million VMT)
Lake	4.296	17	16	395.7
Modoc	4.071	3	3	73.7
Trinity	3.487	4	4	114.7
Calaveras	3.372	9	7	266.9
Inyo	3.294	14	11	425.0
Mono	2.195	6	5	273.3
Del Norte	2.192	4	4	182.5
Sutter	2.169	12	8	553.3
San Benito	1.675	6	6	358.3
Humboldt	1.649	13	10	788.5

Fatality Rate by California County, 2024



The fatality-rate pattern differs substantially from both total crash occurrence and absolute crash burden.

Lake County had the highest fatality rate in 2024, followed by Modoc and Trinity. However, relatively small changes in annual fatalities can produce large changes in rates for counties with limited highway travel, so the exact rankings should be interpreted with some caution.

Repeated High-Rate Counties

- **3 appearances:** Calaveras, Lake, San Benito, Trinity
- **2 appearances:** Del Norte, Plumas, San Francisco, Santa Cruz, Tuolumne
- **1 appearance:** Alameda, Humboldt, Inyo, Los Angeles, Mariposa, Modoc, Mono, Sutter

Calaveras, Lake, San Benito, and Trinity appear in the top 10 across all three exposure-adjusted rate measures. Rather than standing out in only one category, these counties consistently show higher crash occurrence or severity relative to highway travel.

Absolute Burden vs. Exposure-Adjusted Rates

County	Total Number of Crashes	Crash Rate	Number of Fatalities	Fatality Rate
Los Angeles	41,235	1.074	218	0.568
Lake	415	1.049	17	4.296
Modoc	67	0.909	3	4.071
Trinity	208	1.813	4	3.487

A central finding of the project is that **absolute burden and exposure-adjusted rates identify different counties**.

Los Angeles carries the largest absolute crash and fatality burden because of its enormous travel volume, but its exposure-adjusted rates are substantially lower than those of several lower-volume counties. Trinity and Modoc illustrate the opposite pattern: relatively small raw counts can produce high rates when travel exposure is low.

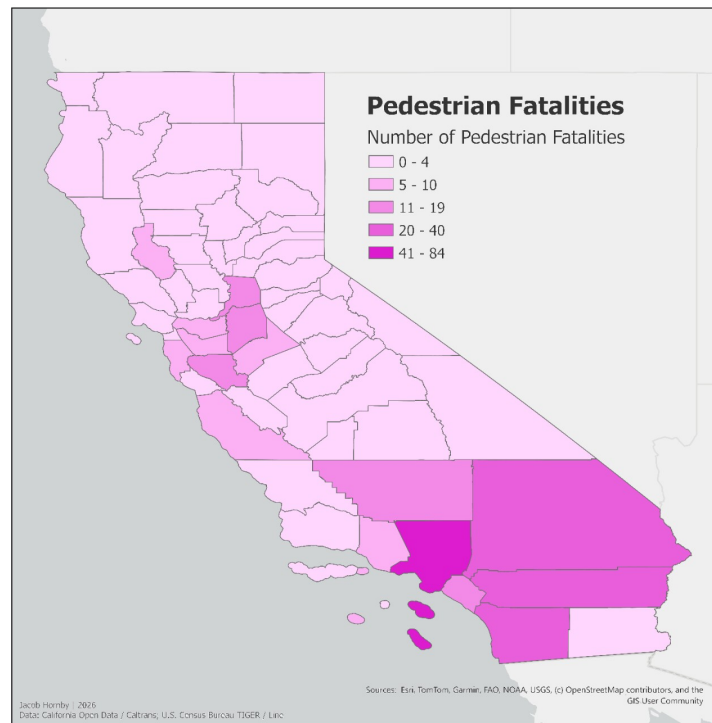
Neither measure is inherently more correct. Raw counts help identify where the largest number of crashes or fatalities occur, while rates help identify where crashes or fatalities are high relative to highway travel.

Pedestrian Fatalities

Top 10 California Counties by Pedestrian Fatalities, 2024

County	Number of Pedestrian Fatalities	Number of Pedestrian Injuries
Los Angeles	84	273
San Diego	40	110
San Bernardino	25	85
Riverside	24	60
Orange	19	104
Sacramento	17	31
Santa Clara	15	39
Kern	14	27
San Joaquin	13	28
Stanislaus	10	28

Pedestrian Fatalities by California County, 2024



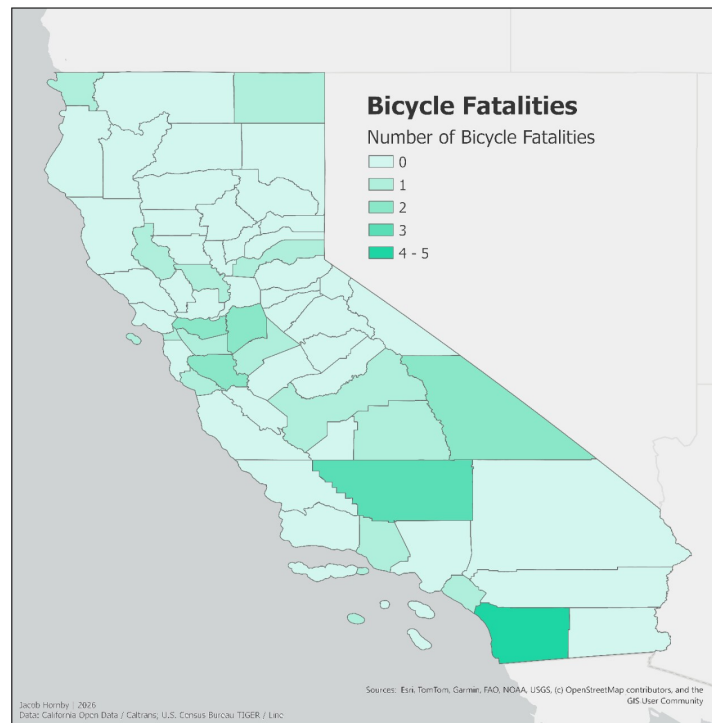
Pedestrian fatalities show a different geographic pattern because they are mapped as raw counts rather than exposure-adjusted rates. The largest burdens are concentrated in high-population and high-traffic counties, particularly in Southern California.

Supplemental Bicycle Fatalities

Top California Counties by Bicycle Fatalities, 2024

County	Number of Bicycle Fatalities	Number of Bicycle Injuries
San Diego	5	34
Kern	3	3
Contra Costa	2	5
Inyo	2	4
Santa Clara	2	21
San Joaquin	2	12
Alameda	1	26
Del Norte	1	1
Fresno	1	5
Lake	1	2

Bicycle Fatalities by California County, 2024



Bicycle fatalities are retained as a supplemental result rather than a primary statewide finding because the distribution is sparse and zero-heavy, with a much narrower range than pedestrian fatalities.

Limitations

Several limitations should guide interpretation:

- This analysis is a one-year snapshot of 2024 conditions. Annual rankings, especially in low-exposure counties, may be sensitive to small changes in crash or fatality counts.
- Exposure-adjusted rates use vehicle travel as the denominator and should not be interpreted as population-based risk.
- Pedestrian and bicycle fatality measures are raw counts and are therefore not directly comparable to the exposure-adjusted crash-rate maps.
- The countywide analytical table does not directly support a rural-versus-urban comparison because its `RURAL_URBAN` field is countywide rather than a county classification.
- County-level aggregation can mask important within-county differences in roadway type, traffic volume, and local conditions.

Conclusion

California's 2024 highway safety patterns depend strongly on how risk and burden are measured. High-volume counties dominate absolute crash and fatality counts, while exposure-adjusted rates elevate a different group of counties.

Across the three primary rate measures, Calaveras, Lake, San Benito, and Trinity repeatedly emerge among the highest-ranking counties. At the same time, counties such as Los Angeles demonstrate that extremely large absolute crash burdens do not necessarily correspond to the highest crash or fatality rates per unit of travel.

Taken together, the results show why both absolute counts and exposure-adjusted rates are useful for highway-safety analysis: they answer different questions and highlight different geographic patterns.

References

- California Department of Transportation (Caltrans). (2026). *2024 Crash Data on State Highway System*. California Open Data.
[2024 Crash Data on State Highway System](#)
- U.S. Census Bureau, Geography Division. (2024). *2024 TIGER/Line Shapefiles: Counties (and Equivalent)*.
[2024 TIGER/Line County Shapefiles](#)

Reproducibility Notes

The files used to reproduce this analysis are available in the project repository:

[California Highway Safety 2024 - GitHub Repository](#)

The repository uses the following structure:

```
california-highway-safety-2024/  
|-- Data/  
|   |-- county-highway-safety-metrics-2024.csv  
|-- Maps/  
|   |-- Total_Crash_Rate_California_2024.png  
|   |-- Fatal_Injury_Crash_Rate_California_2024.png  
|   |-- Fatality_Rate_California_2024.png  
|   |-- Pedestrian_Fatalities_California_2024.png  
|   |-- Bicycle_Fatalities_California_2024.png  
|-- California_Highway_Safety_2024.Rmd  
|-- California_Highway_Safety_2024.pdf  
|-- California_Highway_Safety_2024.html  
|-- README.md  
`-- .gitignore
```

The county-level CSV used here was prepared during the earlier EDA workflow. ArcGIS Pro was used for the spatial join, choropleth design, and final exported map layouts.